

Rishabh Sarma | Curriculum Vitae

Department of Mathematics, 309 McAllister Building
The Pennsylvania State University, University Park, PA-16802, US
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Education

The Pennsylvania State University **State College, United States**
S. Chowla Research Assistant Professor
Mentor : [Ae Ja Yee](#) 2025-2028

University of Florida **Gainesville, United States**
PhD in Mathematics
GPA – 4.00/4.00 2019-2025
Thesis : Modular symmetries for rank-type partition statistics
Advisor : [Frank Garvan](#)

Université Paris Cité **Paris, France**
Chateaubriand Fellow
Advisor : [Jeremy Lovejoy](#) Fall 2023 (Aug-Dec)

Indian Institute of Technology Guwahati **India**
Master of Science in Mathematics and Computing
Masters thesis : Construction of modular forms from theta functions 2016-2018
Advisor : [Anupam Saikia](#)

Tezpur University **India**
Bachelor of Science in Mathematics (with distinction) 2013-2016

Research interests

Partitions, q -series, modular forms, special functions, Ramanujan mathematics, experimental mathematics

Publications and preprints

- Rank and crank difference identities modulo p for $p \equiv 3 \pmod{4}$ (with Frank Garvan), **in preparation**
- Euler's Pentagonal Number Theorem type results via minimal excludants of partitions and overpartitions (with Arित्रam Dhar, Rajat Gupta and Avi Mukhopadhyay), **submitted**
- Congruences modulo 3 for sum of certain parts in overpartitions, **submitted**
- Sum of parts in overpartitions and partitions without repeated odd parts (with Frank Garvan), **submitted** [\[arXiv\]](#)
- Symmetries for overpartition rank and crank functions (with Frank Garvan), **submitted**
- Overpartitions with repeated smallest non-overlined part (with Amita Malik), *Int. J. of Number Theory* (2026), doi:10.1142/S1793042126501411 [\[DOI\]](#)
- Further analogues of Andrews' spt-function identity, *Bull. Austral. Math. Soc.* (2026), 1–10, doi:10.1017/S0004972726101348 [\[DOI\]](#)
- On new minimal excludants of overpartitions related to some q -series of Ramanujan (with Arित्रam Dhar and Avi Mukhopadhyay), *Bull. Austral. Math. Soc.*, **112(1)** (2025), 1–13 [\[DOI\]](#) [\[arXiv\]](#)
- Bailey pairs, radial limits of q -hypergeometric false theta functions, and a conjecture of Hikami (with Jeremy Lovejoy), *Kyushu J. Math.*, **79** (2025), no. 2, 269–285 [\[DOI\]](#) [\[arXiv\]](#)
- Transformation and symmetries for the Andrews-Garvan crank function, *Hardy-Ramanujan J.*, **47** (2024), 1–25 [\[DOI\]](#) [\[arXiv\]](#)
- Combinatorial interpretations of cranks of overpartitions and partitions without repeated odd parts (with Frank Garvan), *SIGMA Symmetry Integrability Geom. Methods Appl.*, **SIGMA 20** (2024), 097, 14 pp [\[DOI\]](#) [\[arXiv\]](#)
- New symmetries for Dyson's rank function (with Frank Garvan), *Ramanujan J.*, **65** (2024), 1883–1939 [\[DOI\]](#) [\[arXiv\]](#)

13. Generalization of the extended minimal excludant of Andrews and Newman (with Arित्रम Dhar and Avi Mukhopadhyay), *J. Integer Seq.*, **26** (2023), Article No. 23.3.7 [DOI] [arXiv]
14. *Solution to* An application of the Jacobi Triple Product, proposed by G. E. Andrews and M. Merca, Problems and Solutions, *The American Mathematical Monthly*, **130**(8) (2023), 765–778 [DOI]
15. On a Euler-Ramanujan formula, Dirichlet series and minimal surfaces (with Rukmini Dey and Rahul Kumar Singh), *Proc.Indian Acad.Sci.(Math. Sci.)*, **130** (2020), Article No. 61 [DOI] [arXiv]

Awards

- **CLAS Dissertation Fellowship**, College of Liberal Arts and Sciences, University of Florida, Summer 2025
- **Chateaubriand Fellowship**, CNRS and Embassy of France in United States, Fall 2023
- **Chat Yin Ho Award** for outstanding graduate student of the year, Department of Mathematics, University of Florida, 2022
- **Mitacs Globalink Research Award** - for research in Canada, 2018
- **Summer Research Fellowship** of the Indian Academy of Sciences, 2015 & 2016
- **INSPIRE Scholarship** (awarded by Department of Science and Technology, Government of India), 2013-18
- **1st rank** in the Assam State Board examination of 12-th grade, 2013

Research visits

University of Cologne, Germany

Visiting Student

Host : Kathrin Bringmann

December 15–20, 2023

International Centre for Theoretical Sciences, TIFR, Bangalore, India

Long Term Visiting Student

Research topic : Dirichlet series and Ramanujan identities

Advisor : Rukmini Dey

Aug, 2018–April, 2019

Simon Fraser University, Burnaby, Canada

SFU-IIT India Initiative Research Intern

Project topic : On a conjecture classifying Littlewood cyclotomic polynomials

Advisor : Stephen Choi

May–July, 2018

Indian Statistical Institute, New Delhi

Summer Research Fellow

Project Topic : Recurrence sequences and primitive divisors and their application in solving Diophantine equations

Advisor : Shanta Laishram

June–July, 2016

Indian Statistical Institute, Bangalore

Summer Research Fellow

Project topic : A study on Spectral theorem for normal matrices

Advisor : Jaydeb Sarkar

June–July, 2015

Talks and presentations

Joint Mathematics Meetings (JMM) 2027, Chicago, IL*

AMS Special Session on Partition Theory, q-Series and Related Topics

"Sum of parts in partitions"

12–15 Jan, 2027

International Congress of Mathematicians (ICM) 2026, Pennsylvania Convention Center, Philadelphia, PA

Poster presentation session

"Modular symmetries and identities for rank-type partition statistics"

23–30 July, 2026

SIAM Annual Meeting, Cleveland, OH

Mini-symposium on Special Functions with Applications in Number Theory and Combinatorics 6-10 July, 2026

"Further analogs of Andrews' spt-function identity"

Special Functions and Number Theory seminar, India (held online)

"Modular symmetries for rank and crank statistics"

18 June, 2026

Partitions and Related Topics I (PART-I), INYAS, India (held online) "Partition and overpartition rank and crank difference identities modulo p for $p \equiv 3 \pmod{4}$ "	26–28 June, 2026
2026 Gainesville International Number Theory conference, University of Florida <i>Conference in honor of Krishna Alladi's 70th birthday</i> "Overpartition rank and crank differences in terms of Dedekind eta-quotients"	18-22 March, 2026
Partitions/combinatorics seminar, Department of Mathematics, Penn State "Analogues of Andrews' spt-function identity"	28 Oct, 2025
AMS Fall Southeastern Sectional Meeting, Tulane University, New Orleans, LA <i>AMS Special Session on Experimental Mathematics</i> "Finding identities for rank and crank of partitions and overpartitions using Maple"	3-5 Oct, 2025
Algebra and Number Theory seminar, Department of Mathematics, Penn State "Modular symmetries for rank-type partition statistics"	23 Sept, 2025
Partitions/combinatorics seminar, Department of Mathematics, Penn State "Minimal excludants of partitions and overpartitions"	23 Sept, 2025
Seminar in Partition Theory, q-series and Related Topics, Michigan Tech (held online) "Bailey pairs and radial limits of q -hypergeometric false theta functions"	6 Feb, 2025
Joint Mathematics Meetings (JMM) 2025, Seattle, WA <i>AMS Special Session on Partition Theory and q-series</i> "New symmetries for overpartition rank and crank functions"	8-11 Jan, 2025
Seminar on Number Theory, Tezpur University (held online) "Bailey pairs and radial limits of q -hypergeometric false theta functions"	19 Nov, 2024
AMS Fall Central Sectional Meeting, University of Texas at San Antonio, TX <i>AMS Special Session on Additive Number Theory and Modular Forms</i> "Cranks of overpartitions and partitions without repeated odd parts – combinatorial interpretations and mock theta connections"	14-15 Sept, 2024
The Legacy of Ramanujan conference, Penn State University, PA <i>Poster presentation session</i> "Combinatorial interpretations and mock theta connections of two residual cranks of overpartitions"	6-9 June, 2024
Number Theory seminar, Department of Mathematics, University of Florida "Bailey pairs and radial limits of q -hypergeometric false theta functions"	9 April, 2024
International conference on modular forms and q-series, University of Cologne <i>Speed talk session</i> "Pursuing Dyson's dream - unraveling new symmetries for rank-type statistics"	11-15 March, 2024
Joint Mathematics Meetings (JMM) 2023, Boston, MA <i>AMS Special Session on Quadratic Forms, Modular Forms, and Applications</i> "New symmetries for the Andrews-Garvan crank function"	4-7 Jan, 2023
SASTRA-Ramanujan Award ceremony and conference, Kumbakonam, India "New symmetries for rank and crank generating functions"	20-22 Dec, 2022
AMS Fall Central Sectional Meeting, University of Texas at El Paso, TX <i>AMS Special Session on The Intersection of Number Theory and Combinatorics</i> "Relating the smallest missing part to other partition theoretic objects and statistics"	17-18 Sept, 2022
Number Theory seminar, Department of Mathematics, University of Florida "Transformation and symmetries for Dyson's rank function"	13 Sept, 2022
Student Symposium in Combinatorics, UNSW (held online) "Relating the smallest missing part to other partition statistics"	30 May–3 June, 2022
47th Annual NY State Graduate Mathematics Conference, Syracuse University	2 April, 2022

“New symmetries for Dyson’s rank function”

Masters seminar, Department of Mathematics, Indian Institute of Technology Guwahati

“Cyclotomic polynomials and the Bunyakovski Conjecture”

27 Oct, 2017

Conferences and workshops attended

The Gross-Zagier Formula, 40 Years Later

Massachusetts Institute of Technology, Cambridge, MA

3-7 Aug, 2026

Joint Mathematics Meetings (JMM) 2022

held virtually

6-9 April, 2022

2020 AMS Fall Eastern Sectional Meeting

held virtually, formerly at The Pennsylvania State University

3-4 Oct, 2020

2019 AMS Fall Southeastern Joint Sectional Meeting

University of Florida, Gainesville

2-3 Nov, 2019

Advanced Instructional School on Modular forms

Indian Institute of Technology Guwahati

13 May-1 June, 2019

Algebraic and analytic aspects of automorphic forms

International Centre for Theoretical Sciences, TIFR, Bangalore

25 Feb-7 March, 2019

Annual Foundation School - I

Indian Institute of Technology Guwahati

Dec 1-31, 2016

Advanced topics on Discrete Logarithms over Finite Fields and Elliptic Curves

Indian Statistical Institute, New Delhi

14-22 July, 2016

26th Mid-Year Meeting of scientific lectures of the Indian Academy of Sciences

Indian Institute of Science, Bangalore

3-4 July, 2015

National Board of Higher Mathematics Workshop on Algebra and Analysis

Tezpur University

17-23 Feb, 2015

Asian Science Camp 2014

Nanyang Technological University, Singapore

24-29 Aug, 2014

Referee work

Refereed articles for **International Journal of Number Theory**, **Journal of Algebraic Combinatorics**, **The Journal of the Indian Mathematical Society**, **The Ramanujan Journal** and **Mathematical Reviews (MathSciNet)**

Teaching experience

The Pennsylvania State University.....

- Instructor for **Matrices** ([MATH 220](#)), Fall 2026
- Instructor for **Ordinary Differential Equations** ([MATH 250](#)), Summer 2026 (online)
- Instructor for **Matrices** ([MATH 220](#)), Spring 2026
- Instructor for **Calculus with Analytic Geometry II** ([MATH 141](#)), Spring 2026 & Fall 2025

University of Florida.....

- Teaching assistant for **Elementary Differential Equations** (MAP 2302), Spring 2025
- Teaching assistant for **Differential Equations for Engineers and Physical Scientists** (MAP 4305) & **Intermediate Differential Equations** (MAP 5304), Spring 2024
- Teaching assistant for **Calculus III** (MAC 2313), Summer 2024 & Fall 2022

- Teaching assistant for **Calculus II** (MAC 2312), Summer 2021 & Spring 2021
- Teaching assistant for **Calculus I** (MAC 2311), Fall 2022, Fall 2020, Spring 2020 & Fall 2019
- Teaching assistant for **Survey of Calculus II** (MAC 2234), Fall 2021
- Teaching assistant for **Survey of Calculus I** (MAC 2233), Fall 2024
- Co-instructor for **Precalculus Algebra** (MAC 1140), Summer 2020
- Instructor for **Precalculus Algebra** (MAC 1140), Summer 2022
- Teaching assistant for **Precalculus Algebra** (MAC 1140), Spring 2022
- Co-instructor for **Basic College Algebra** (MAC 1105), Summer 2023 & Spring 2023

Professional service

- Co-organizer of [Combinatorics/Partitions Seminar](#) in the Department of Mathematics, The Pennsylvania State University, Spring 2026 - present
- Co-organizer of AMS Special Session on Partitions, q -series and modular forms with Ae Ja Yee at the [2026 AMS Fall Eastern Sectional Meeting](#), Department of Mathematics, George Washington University, Washington, DC, October 3-4, 2026*
- Assistant organizer of [2026 Gainesville International Number Theory conference \(Alladi70\)](#) in honor of Krishna Alladi's 70th birthday, Department of Mathematics, University of Florida, Gainesville, FL, March 18 - 22, 2026

Graduate coursework

Modern Algebra, Modern Analysis, Functional Analysis, Complex Analysis, Numerical Analysis, Theory of Partitions and q -series, Modular Forms and Mock-Theta Functions, Topics in Analytic Number Theory and Sieve Theory, Topics in Combinatorial Theory, Galois Theory, Algebraic Topology, Differential Topology, Probability and Potential Theory

Computer skills

C, C++, $\text{\LaTeX}$, Maple, Python, SageMath

References

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*Upcoming